

CaseStudy 1: Customer Churn Risk & Retention Optimization (DSPristine)

Case Study – Dyashin Technosoft Pvt Ltd

Company Overview

Dyashin Technosoft Pvt Ltd is an IT Services organization providing software development, IT consulting, training, and digital solutions to clients across multiple regions.

Domain

Human Resources Analytics (HR Tech)

Project Duration

1 Month

Business Scenario

Dyashin Technosoft has developed an HR management platform called **DSPristine** used by organizations to manage employee records and HR operations.

One of the biggest challenges faced by HR teams is **employee attrition**. Losing skilled employees leads to productivity loss and increased hiring costs.

To enhance DSPristine, Dyashin plans to integrate a **Customer Churn Risk & Retention Optimization module** that analyses employee behaviour and predicts which employees are likely to leave the company.

By identifying employees at risk of leaving, HR teams can take preventive actions such as salary adjustments, role changes, or engagement programs.

Business Objectives

- Identify employees at high risk of leaving the organization
 - Reduce employee turnover rates
 - Identify key factors contributing to employee attrition
 - Enable HR teams to take proactive retention measures
 - Improve employee satisfaction and engagement
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Project Scope

- Handling class imbalance in attrition datasets
 - Feature engineering using employee behaviour and tenure
 - Exploratory data analysis to identify attrition patterns
 - Building classification models
 - Evaluating prediction performance
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Models Developed

- Employee Attrition Predictor
 - Retention Risk Analyzer
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Expected Deliverables

- Python Notebook implementing the predictive model
 - Null values will be handled using imputation techniques (mean/mode/median) instead of dropping.
 - Duplicate records will be analysed and handled appropriately, not blindly removed.
 - Inconsistent categorical values will be standardized.
 - Incorrect data types will be corrected.
 - Outliers will be treated using statistical methods.
 - Visualization report showing key attrition drivers.
 - Model evaluation report including key performance metrics such as Accuracy, Precision, Recall, and F1 Score to assess prediction effectiveness.
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Expected Outcome

This AI integration enhances **DSPristine**, transforming it into an intelligent HR platform capable of predicting employee attrition and improving retention strategies.